

meets the World Health Organization's air quality benchmark for PM 2.5 at 5 micrograms per cubic meter far stricter than India's 40", says Rajiv Mangal, Vice President, Safety, Health & Sustainability, Tata Steel. The firm applies this foresight domestically by adopting advanced pollution control technologies such as high-efficiency bag filters over traditional electrostatic precipitators. This builds long-term credibility and operational resilience in an increasingly environmentally conscious marketplace, adds Mangal.

The global shift towards carbon pricing and border adjustments also shapes Tata Steel's sustainability roadmap. The European Union's Carbon Border Adjustment Mechanism is a clear signal of where the global steel market is headed. With Europe imposing carbon taxes of €60–70 per tonne of CO₂, CBAM seeks to level the playing field by ensuring imported steel meets similar environmental stan-

dards. "For Tata Steel's Netherlands operations, this corrects a historical disadvantage against high-emission steel imports. More broadly, Tata Steel advocates a forward-thinking model for India, proposing that carbon taxes collected on exports be reinvested domestically to fund decarbonization efforts. While this proposal is yet to gain traction internationally, it offers a potential blueprint for developing economies to turn carbon tariffs into catalysts for sustainable development", says Mangal.

Tata Steel's decision-making is shaped not just by regulation but by internal accountability. It uses a shadow carbon price of \$40 per tonne of CO₂ to guide investment and operational choices, even though India lacks a formal carbon tax. This internal pricing signals a strategic alignment with a low-carbon future, sometimes at the cost of short-term profits. In its efforts the company led the world's larg-

est hydrogen injection trial in a blast furnace at its Jamshedpur plant and operates a commercial-scale carbon capture facility building capabilities ahead of regulatory and market shifts.

Water stewardship is a strategic pillar the company focusses on recognizing the steel sector's high water demands and hence has a need for ecological balance. After a 1.5-year study of the Brahmani River basin in Odisha, where its expansion is centered. Tata Steel is committed to becoming water neutral by 2030 and water positive thereafter.

Inclusivity is another cornerstone of Tata Steel's sustainability agenda. Implementation is nuanced by local laws: in Odisha, women can work night shifts; in Jharkhand, home to Jamshedpur, they cannot. Despite such hurdles, Tata Steel is investing in inclusive infrastructure and advocating policy change. It has hired over 100 transgender individuals into operational roles like locomotive and heavy vehicle operators and is additionally, training and recruitment programs for persons with disabilities and underrepresented communities.

Extending sustainability beyond its boundaries, Tata Steel enforces strong ESG (Environmental, Social, and Governance) criteria across its supply chain. Large suppliers are evaluated on emissions and labour practices, while MSMEs receive support to gradually meet these benchmarks. The company monitors Scope 3 emissions across its supply chain and has set reduction targets. It maintains zero tolerance on child labour and human rights violations, with policies that extend to biodiversity and workplace safety.

Looking ahead, Tata Steel sees sustainability not as a peripheral strategy, but a core business imperative. **BT**

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OVERALL SPECIAL JURY AWARD IN SUSTAINABILITY LEADERSHIP

→ With a target of net-zero emissions by 2045, Tata Steel ranks among a small cohort of global steelmakers committed to early climate action

→ The company has adopted advanced pollution control technologies such as high-efficiency bag filters

→ This not only prepares Indian operations for tougher future norms but also builds long-term credibility and operational resilience

→ It uses a shadow carbon price of \$40 per tonne of CO₂ to guide investment and operational choices